



Neutrinos via Charm Decays in Astrophysical Sources

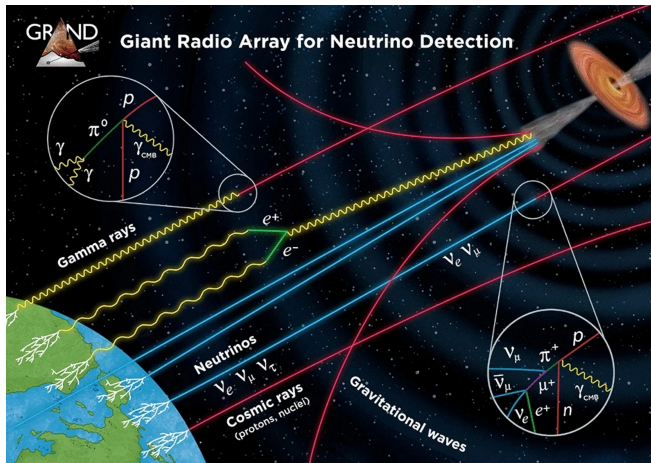
TALK IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE DEGREE OF BACHELOR OF SCIENCE

Fritz Ali Agildere

September 3, 2024

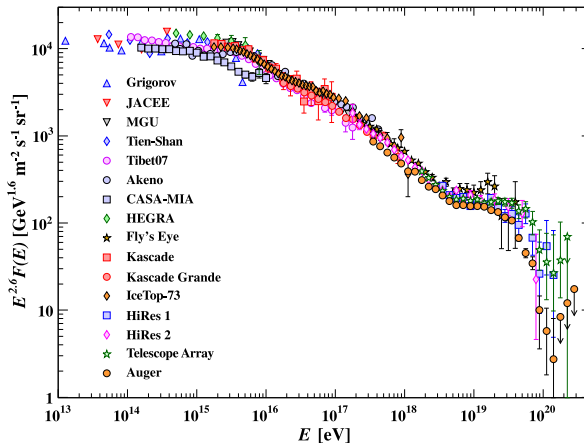
Technical University of Dortmund & Ruhr University Bochum

Multimessenger Astrophysics



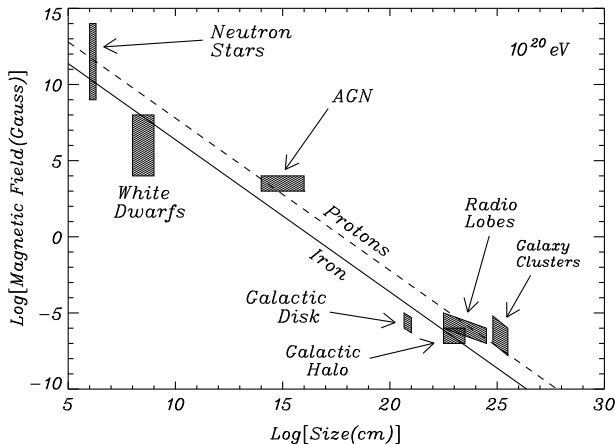
X. L. Qian et al.
Nucl. Sci. Tech. (2021)

Cosmic Ray Spectrum & Cutoff



K. A. Olive et al.
Chin. Phys. C (2014)

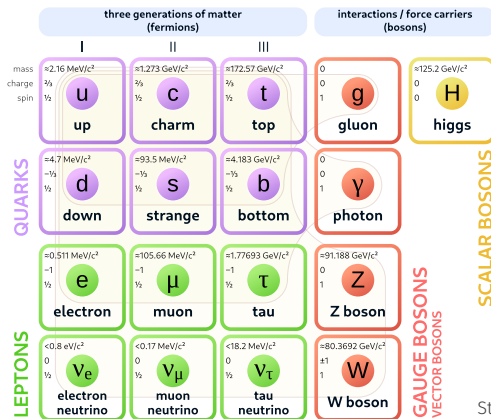
Plausible Cosmic Ray Origins



A. V. Olinto
AIP Conf. Proc. (2000)

Elementary Particles & Hadrons

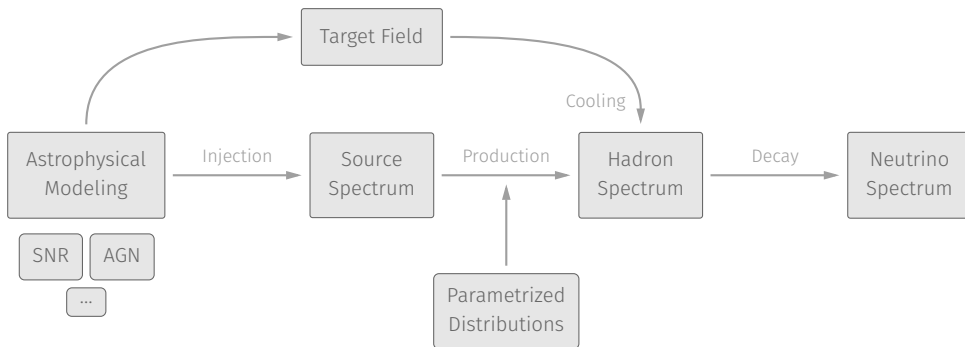
Standard Model of Elementary Particles

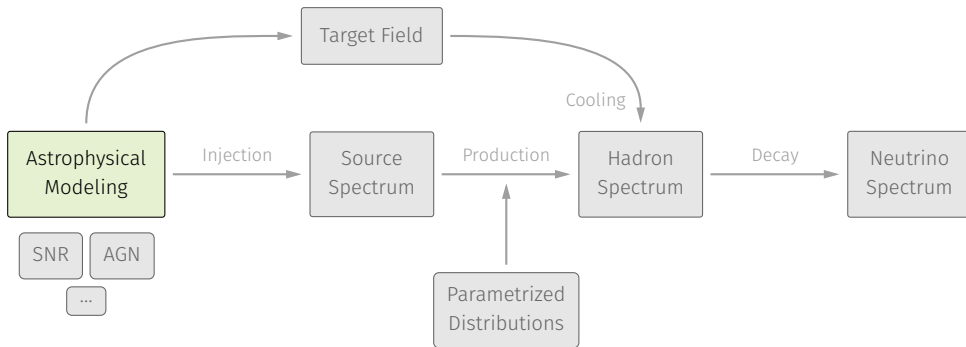


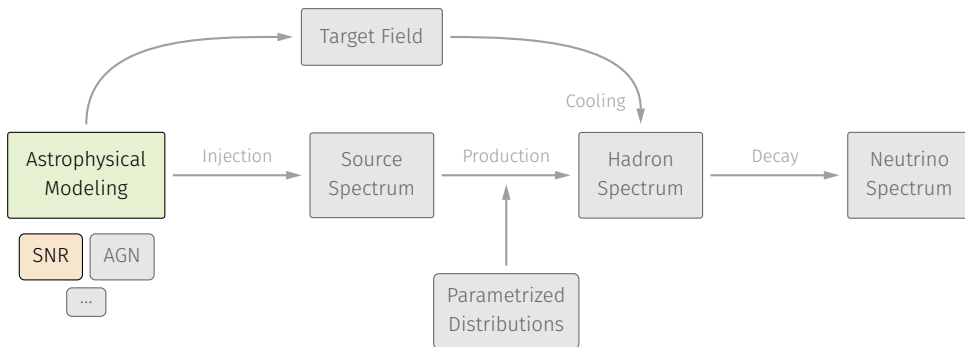
Standard Model
Wikimedia Commons (2019)

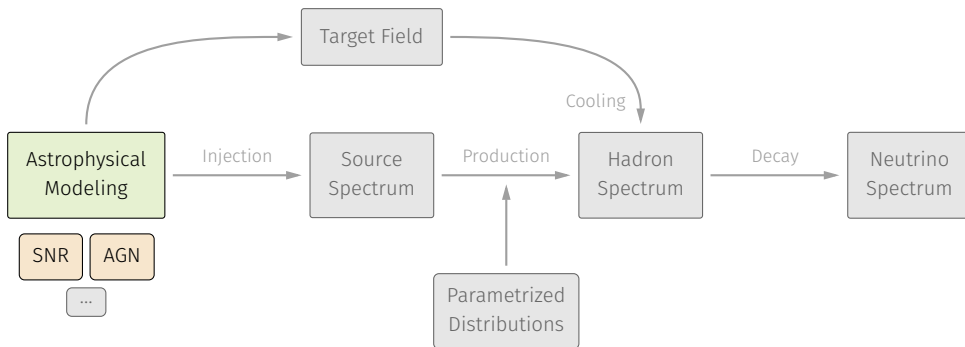
Relevant Hadron Properties

h	q_h	τ_h / s	m_h / GeV	s_h / GeV^2	$B_{h \rightarrow \nu}$
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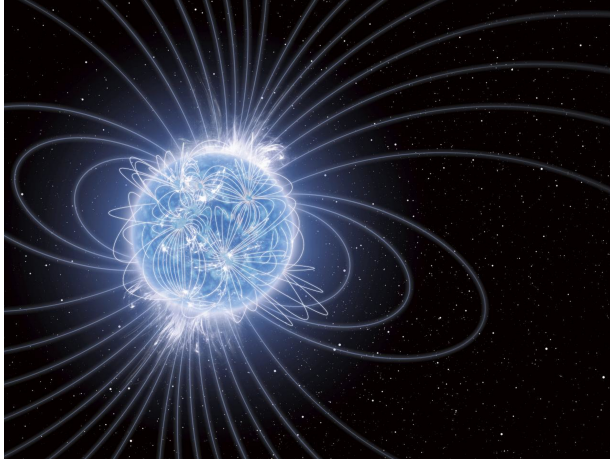






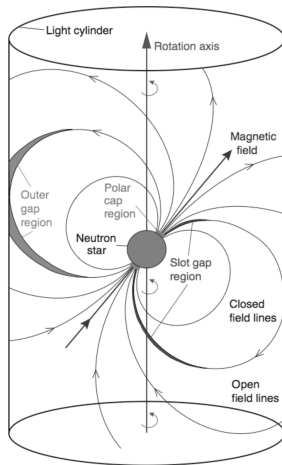


Young Neutron Stars



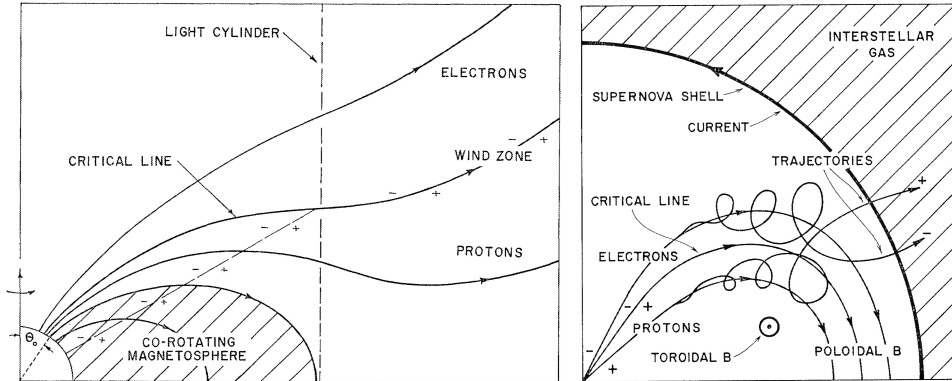
R. Takeshige
Magnetar (2020)

Dipolar Pulsar Model



E. Aliu et al.
Science (2008)

Particle Acceleration



P. Goldreich, W. H. Julian *ApJ* (1969)



Magnetar Spindown

- Electromagnetic Luminosity

Magnetar Spindown

■ Electromagnetic Luminosity

$$L = \frac{B_{ns}^2 R_{ns}^6 \omega_0^4 (1 + \sin^2 \chi)}{4c^3 (1 + t/t_{sd})^2}$$

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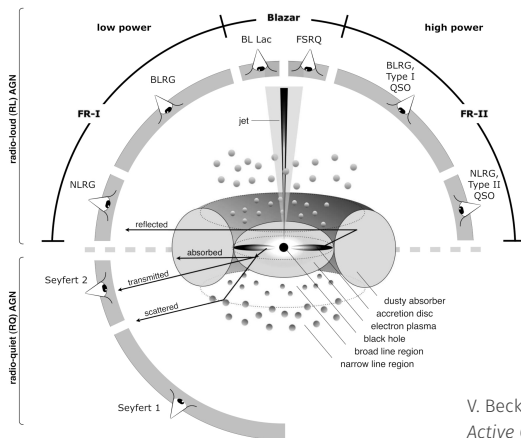
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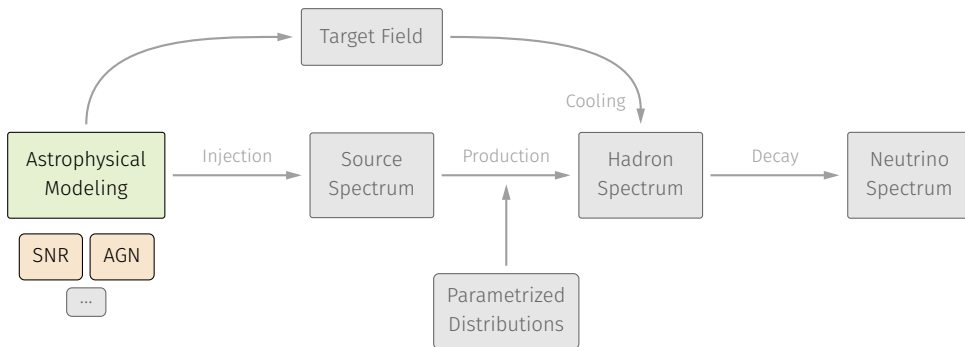
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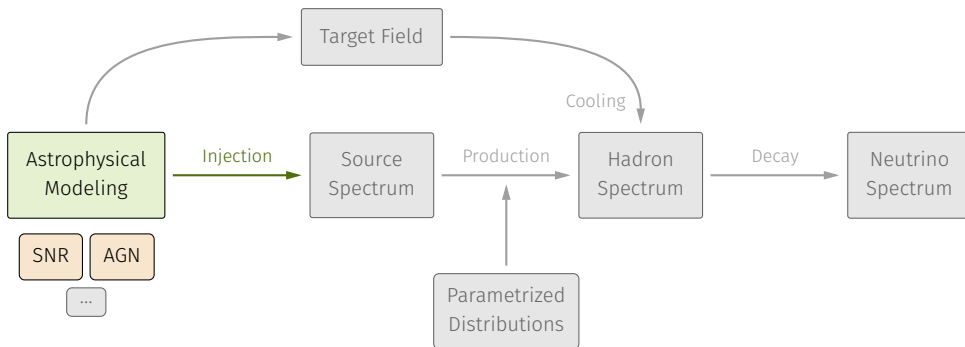
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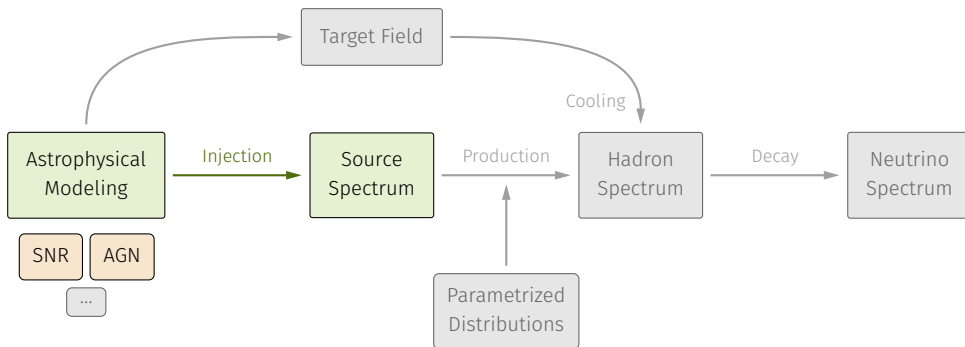
Active Galactic Nuclei



V. Beckmann, C. Shrader
Active Galactic Nuclei (2012)









Injected Primary Spectrum

- Monochromatic Proton Energy

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$$\frac{d\dot{N}_p}{dE_p} = \frac{B_{ns} R_{ns}^3 \omega_0^2}{ec(1 + t/t_{sd})} \delta(E_p - E^M)$$

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Injected Primary Spectrum

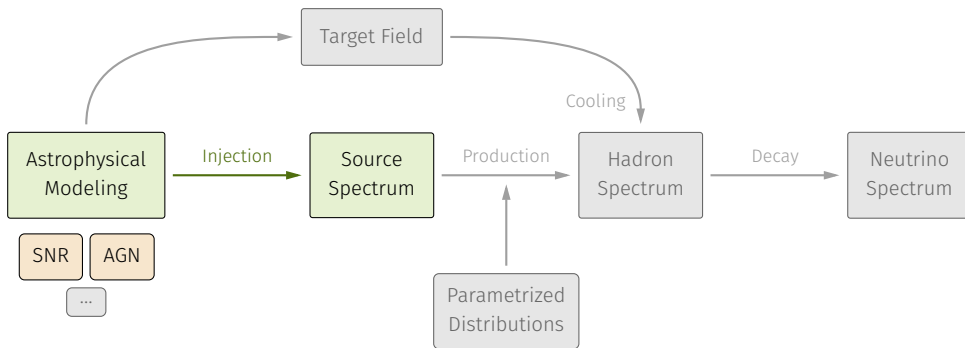
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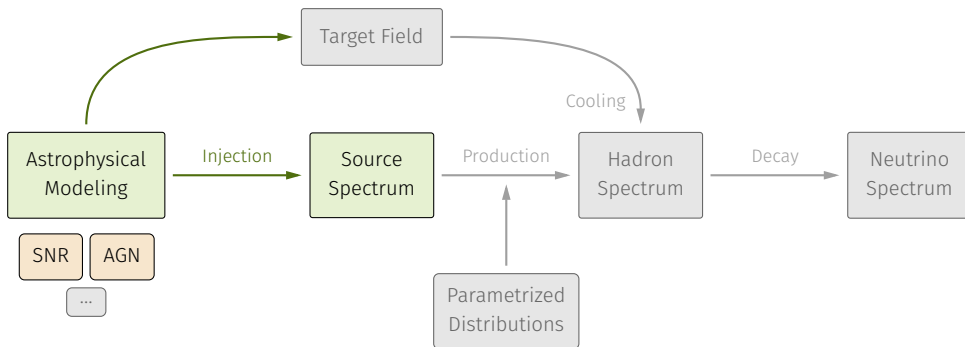
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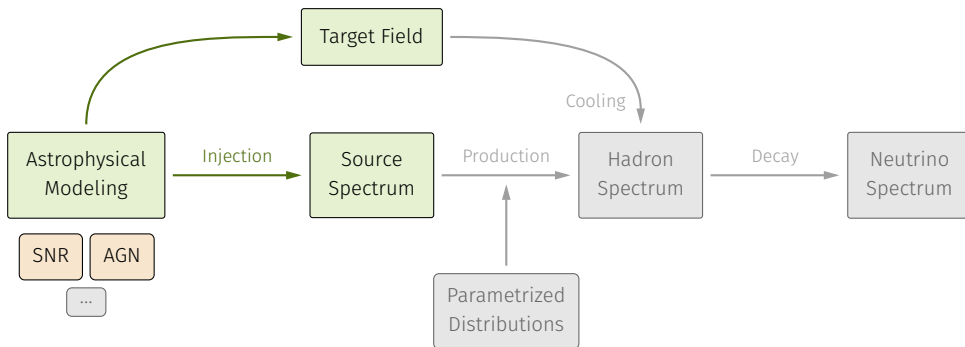
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■ Numerical Setup

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$$\begin{array}{llll} I_{ns} = 10^{45} \text{ g cm}^2 & R_{ns} = 10^6 \text{ cm} & B_{ns} = 10^{14.5} \text{ G} & t_{sd} = 10^{3.5} \text{ s} \\ M_{ej} = 10 M_{\odot} & \beta_{ej} = 0.1 & \eta = 0.1 & \end{array}$$



Accretion Disk

- Thin Limit Scale Height

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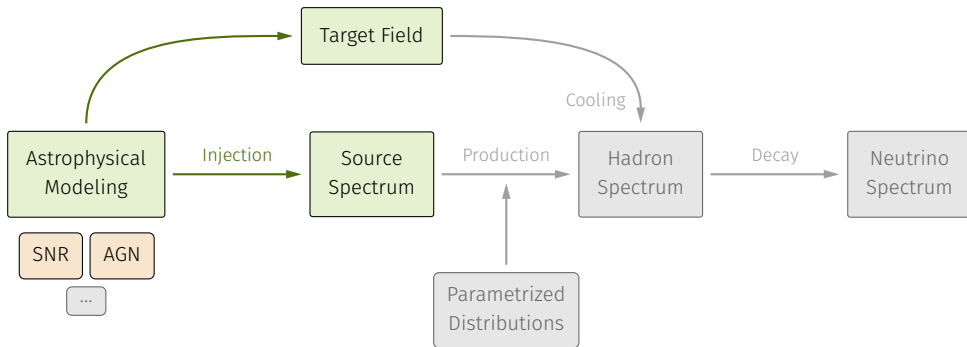
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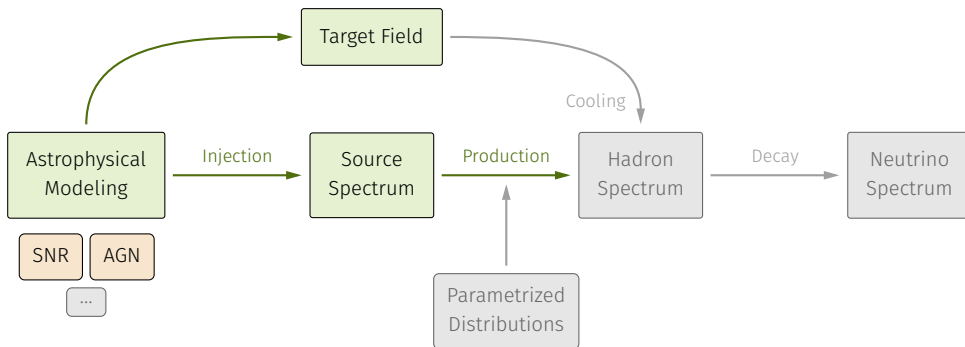
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Direct Hadron Spectrum

- Single Interaction Distribution

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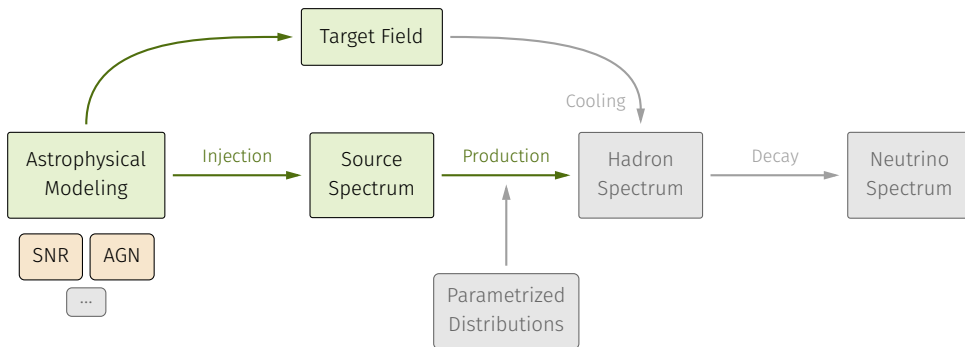
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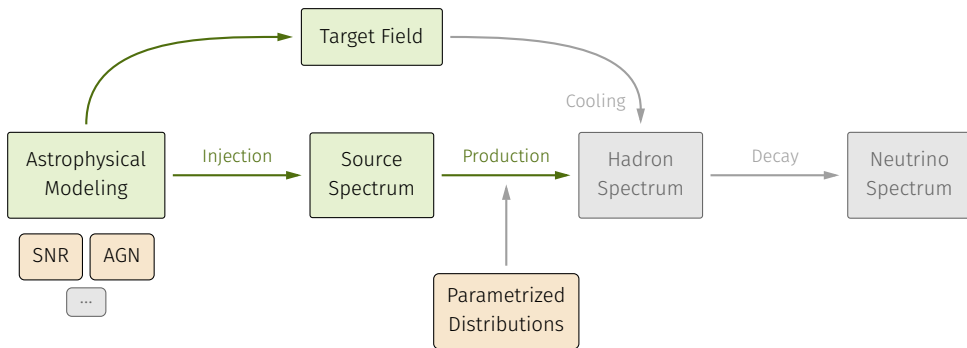
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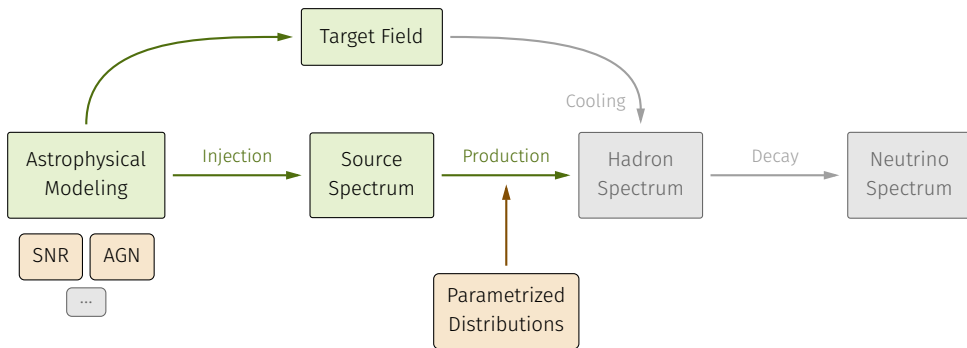
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Unweighted Energy Differential Cross Section

- Fragmentation Function

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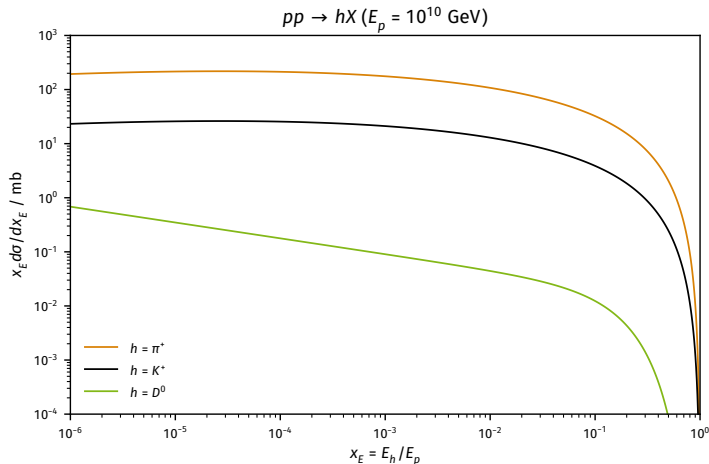
■ Energy Fractions

$$x_E = E_h/E_p$$

$$x_c = E_c/E_p$$

$$z = E_h/E_c$$

Weighted Energy Differential Cross Section

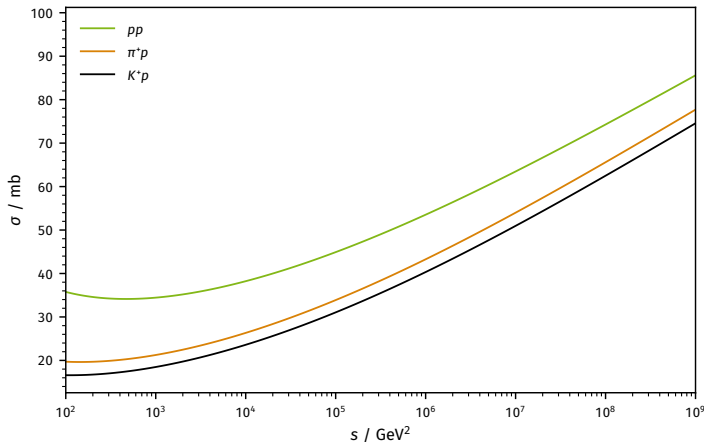


Derived Parametrizations

S. R. Kelner et al.
Phys. Rev. D (2006)

V. P. Gonçalves et al.
JHEP (2007)

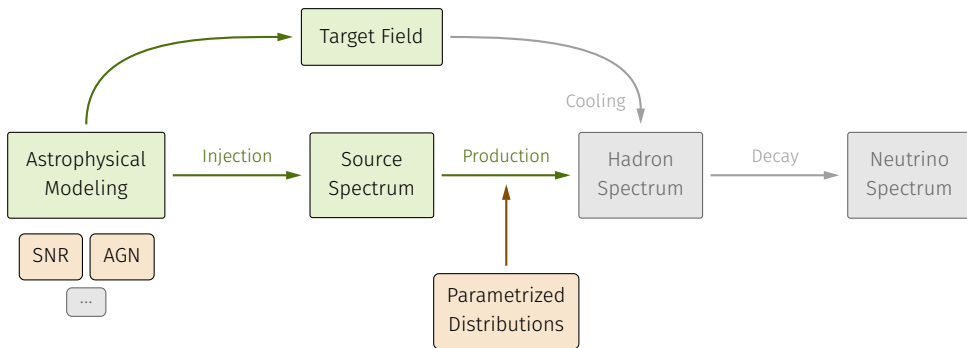
Hadron Inelastic Scattering Cross Section

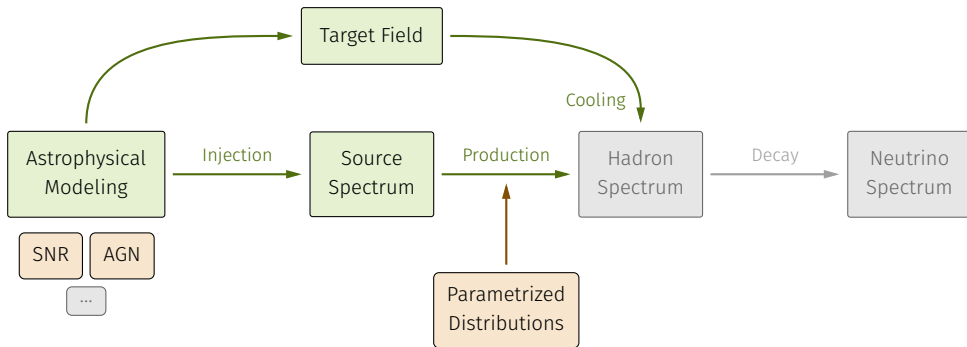


Combined Parametrizations

V. I. Belousov et al.
Phys. Atom. Nuclei (2016)

D. A. Fagundes, M. J. Menon
Nucl. Phys. A (2012)







Hadronic Cooling & Decay

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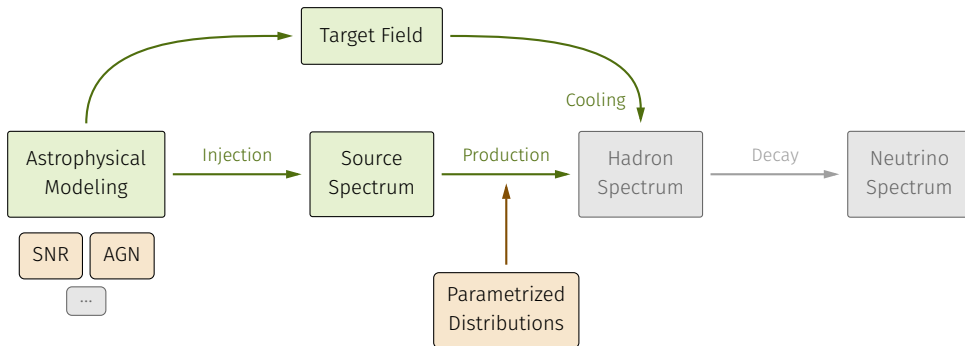
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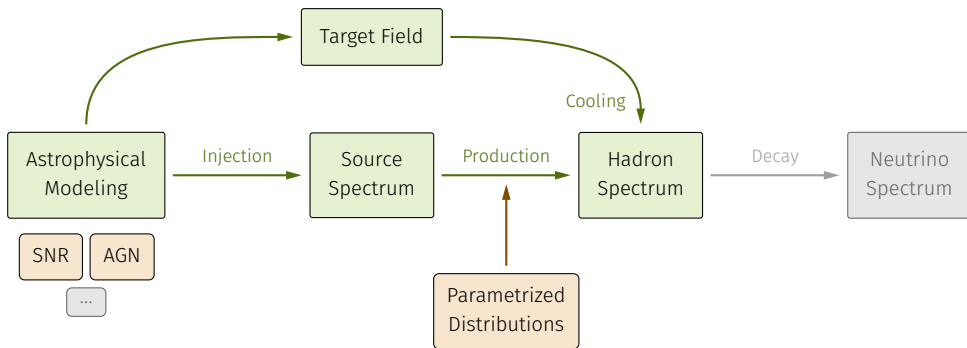
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$$O = \kappa_{pp} \sigma_{pp} n d$$

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$$\frac{dN_h}{dE_h} = CO \int_{E_h}^{\infty} dE_p \frac{dN_p}{dE_p} F_{pp \rightarrow h}$$

■ Temporal Evolution

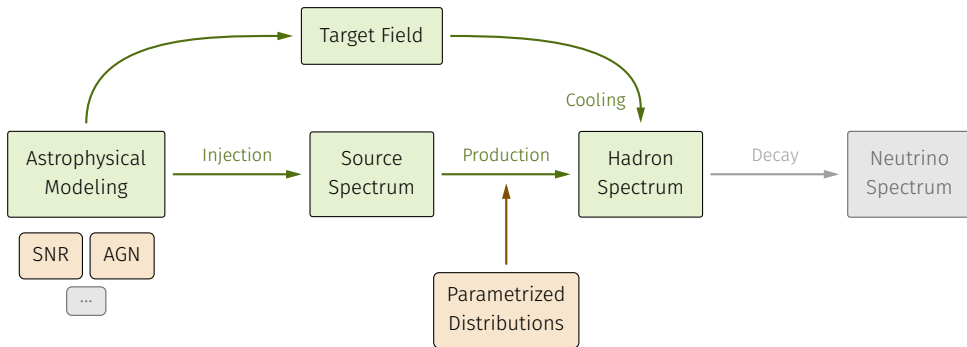
Produced Hadron Spectrum

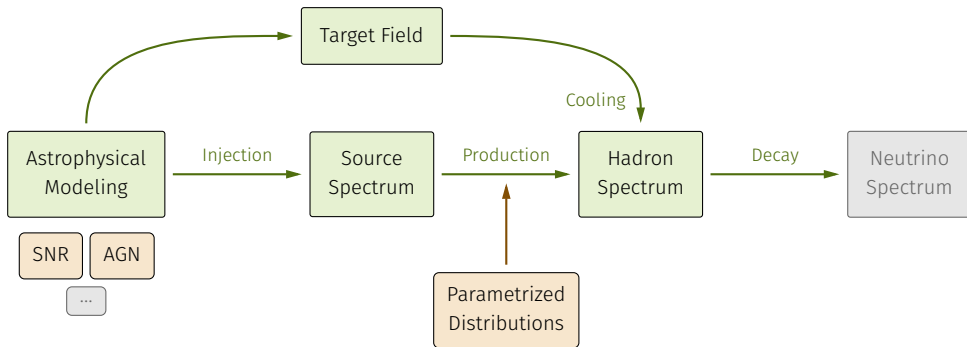
■ Modification Factors

$$\frac{dN_h}{dE_h} = CO \int_{E_h}^{\infty} dE_p \frac{dN_p}{dE_p} F_{pp \rightarrow h}$$

■ Temporal Evolution

$$CO \propto t^{-3} (1 - e^{-t^3})$$







Resulting Neutrino Spectrum

- Complex Three Body Decay

Resulting Neutrino Spectrum

- Complex Three Body Decay
- Simple Two Body Decay

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$$\lambda_h = m_\mu^2 / m_h^2$$

Resulting Neutrino Spectrum

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$$F_{h \rightarrow \nu} = B_{h \rightarrow \nu} E_h^{-1} (1 - \lambda_h)^{-1}$$

Resulting Neutrino Spectrum

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Resulting Neutrino Spectrum

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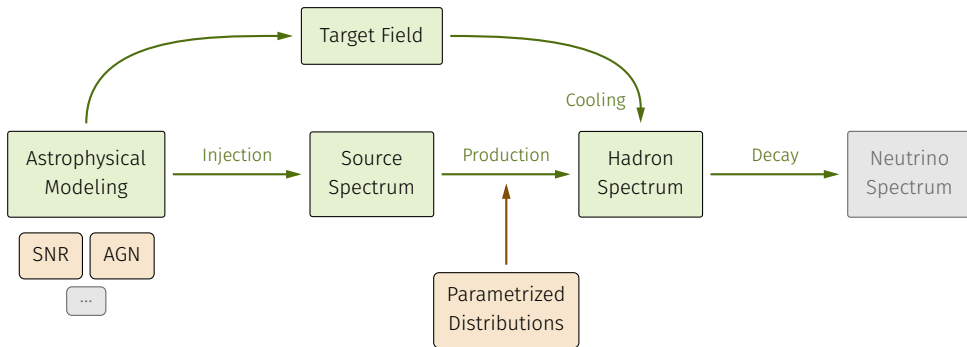
$$F_{h \rightarrow \nu} = B_{h \rightarrow \nu} E_h^{-1} (1 - \lambda_h)^{-1}$$

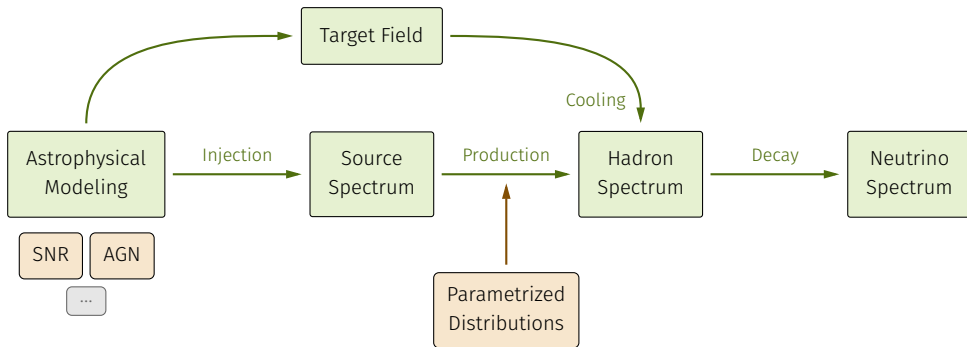
- Hadron Number Convolution

$$\frac{dN_\nu}{dE_\nu} = \int_{E_\nu}^{\infty} dE_h \frac{dN_h}{dE_h} F_{h \rightarrow \nu}$$

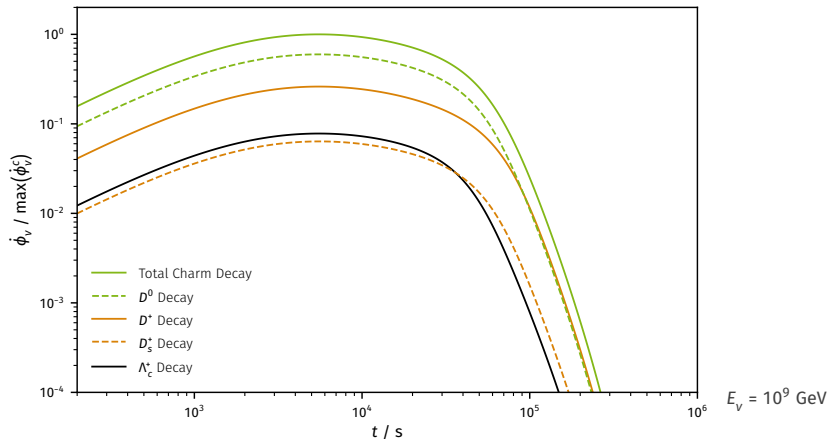
Relevant Hadron Properties

h	q_h	τ_h / s	m_h / GeV	s_h / GeV^2	$B_{h \rightarrow \nu}$
p	uud	∞	0.94	15.98	0.000
π^+	$u\bar{d}$	26.0×10^{-9}	0.14	10.23	1.000
K^+	$u\bar{s}$	12.4×10^{-9}	0.49	12.62	0.636
D_0	$c\bar{u}$	0.4×10^{-12}	1.86	0.45	0.067
D^+	$c\bar{d}$	1.0×10^{-12}	1.87	0.40	0.176
D_s^+	$c\bar{s}$	0.5×10^{-12}	1.97	0.71	0.065
Λ_c^+	udc	0.2×10^{-12}	2.29	1.61	0.045

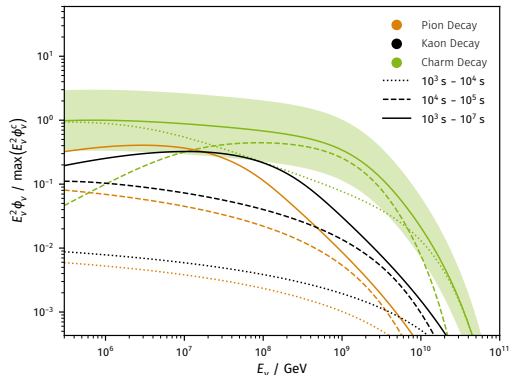
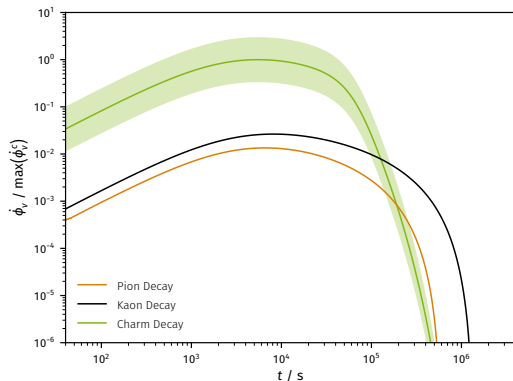




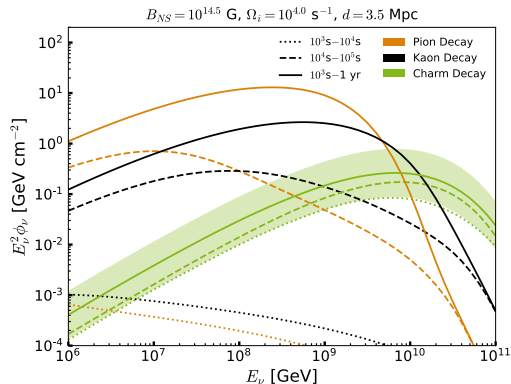
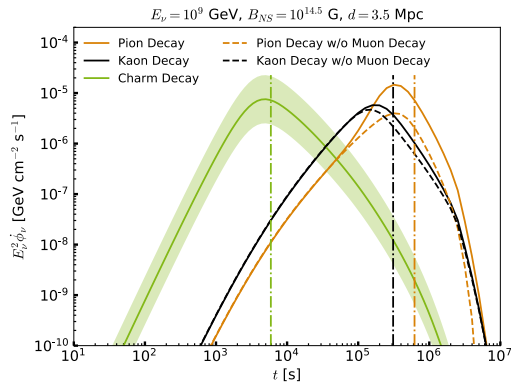
Normalized Magnetar Charm Contributions



Normalized Magnetar Neutrino Flux & Fluence

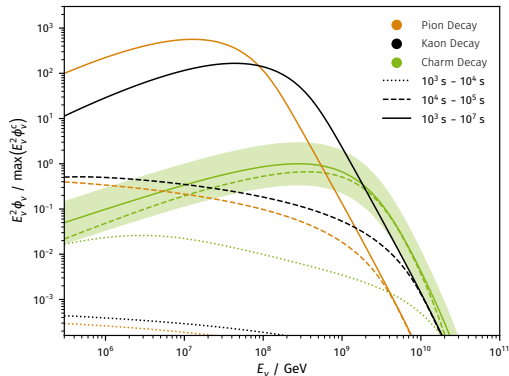
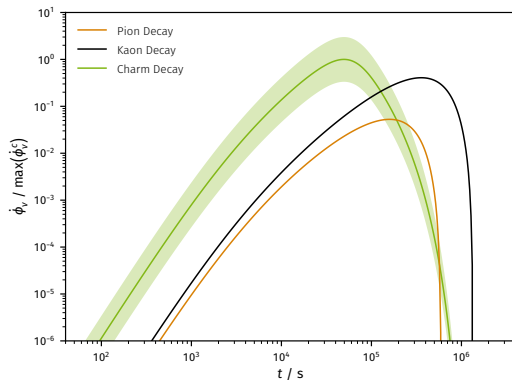


Published Estimated Magnetar Neutrinos

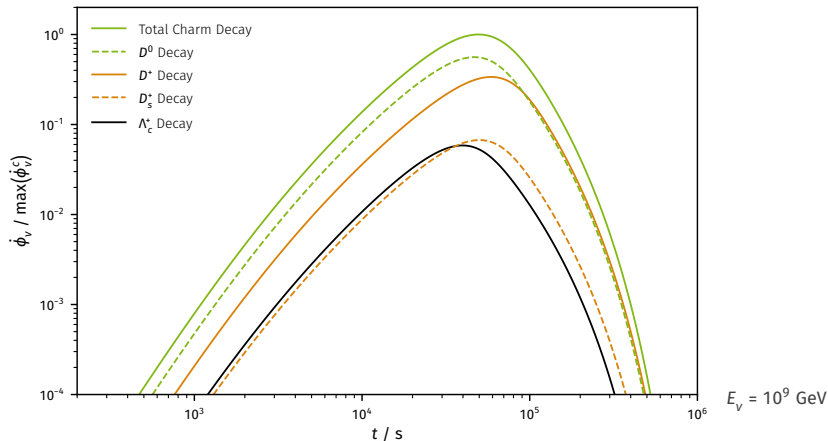


J. A. Carpio et al. *Phys. Rev. D* (2020)

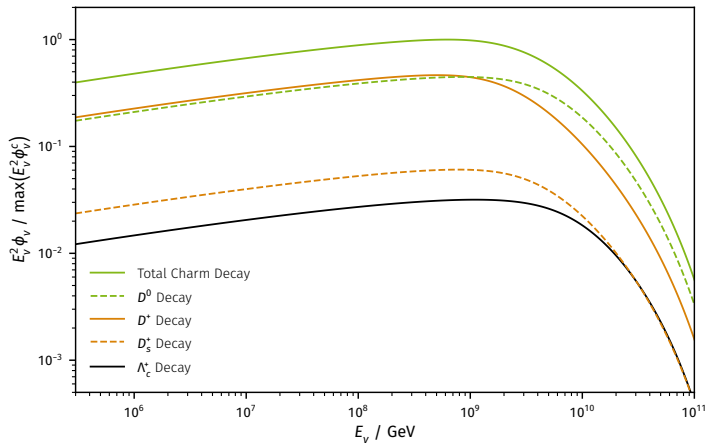
Modified Magnetar Neutrino Flux & Fluence



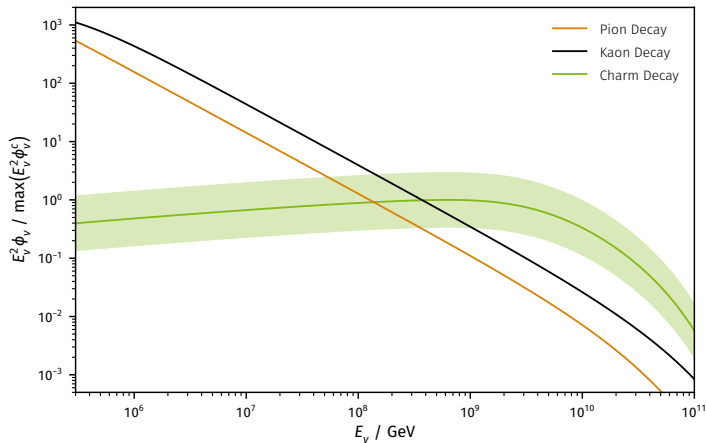
Modified Magnetar Charm Contributions

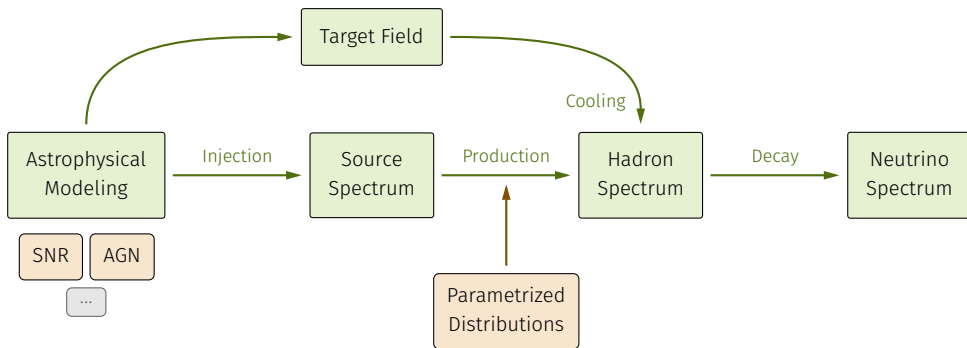


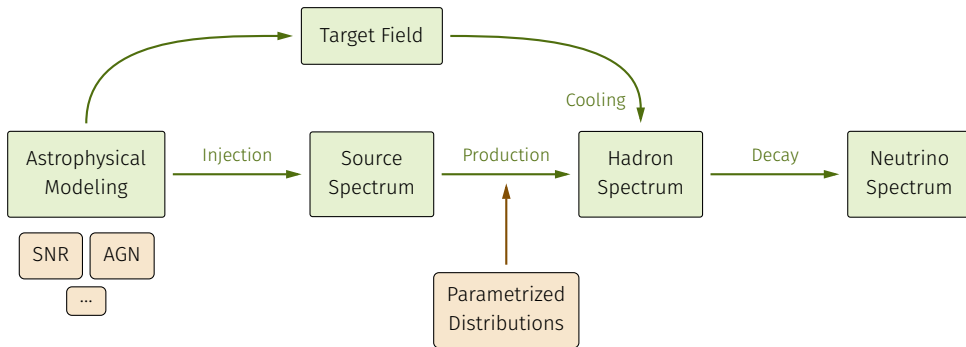
Normalized AGN Charm Contributions



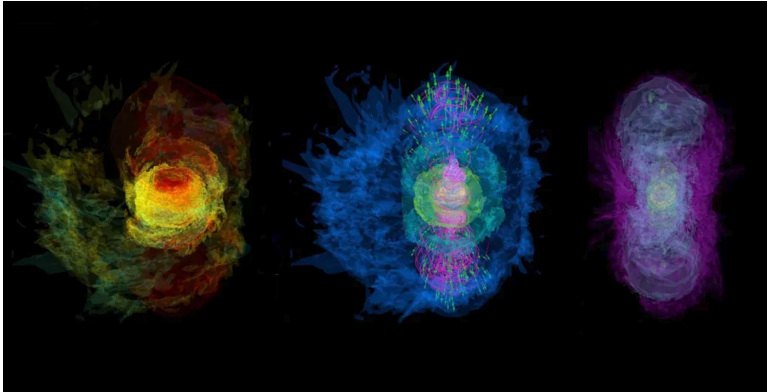
Normalized AGN Neutrino Fluence





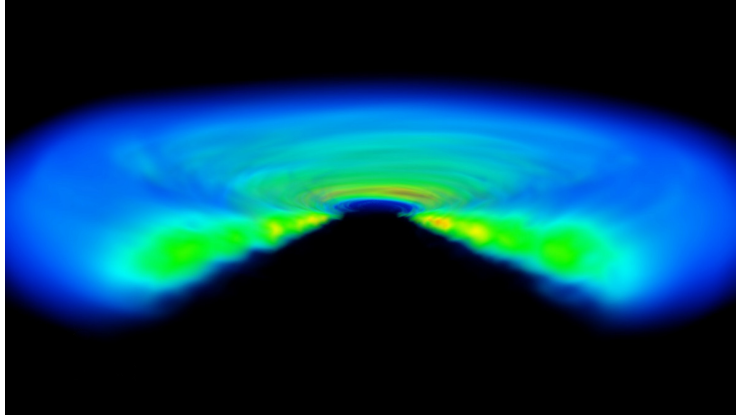


Simulated Magnetar Formation

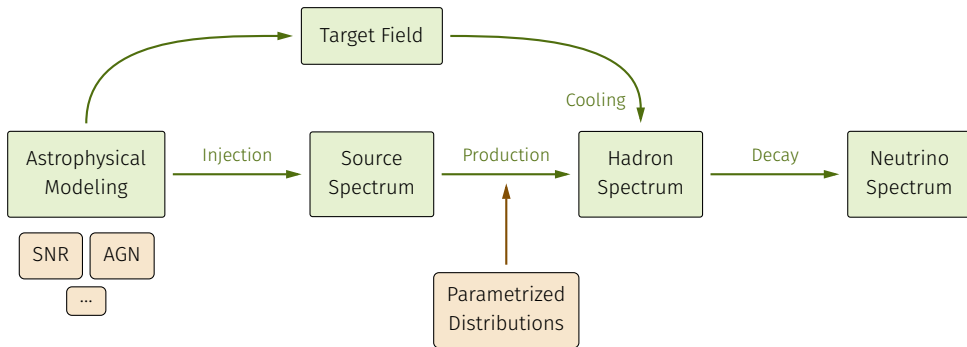


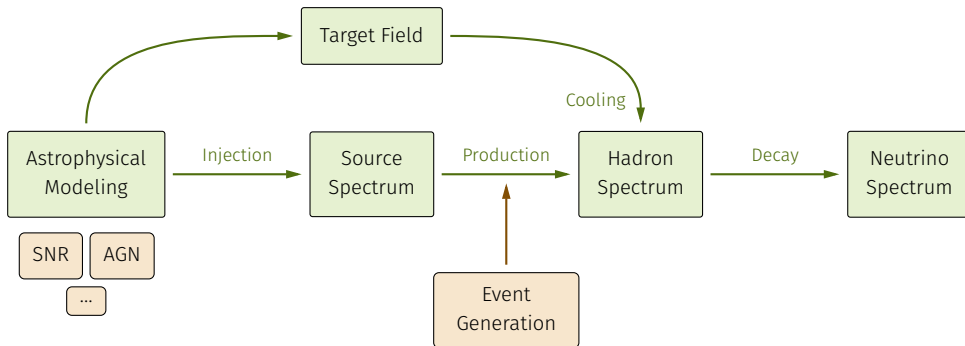
K. Hayashi *MPI Grav. Phys.* (2024)

Accretion Disk Structure

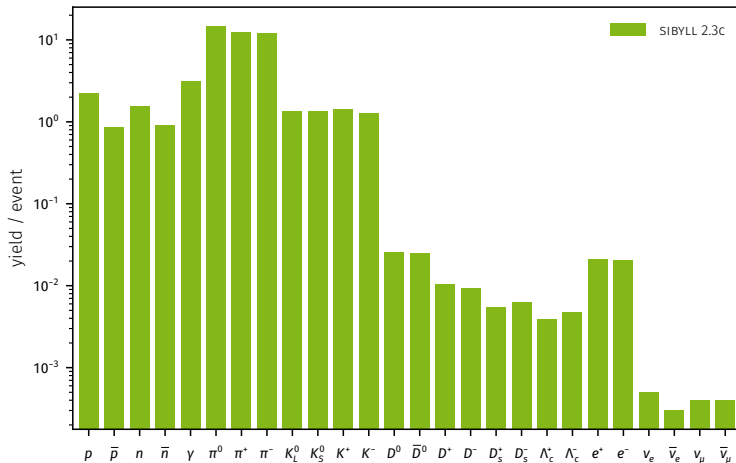


V. V. Zhuravlev et al. *ApJ* (2014)



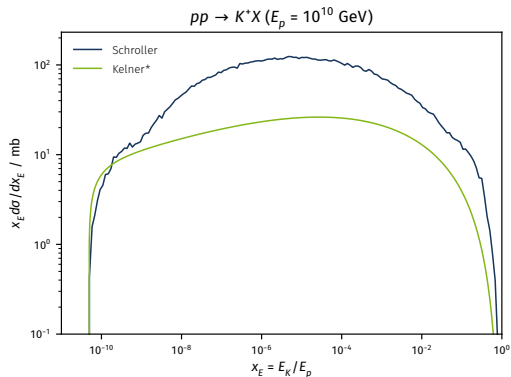
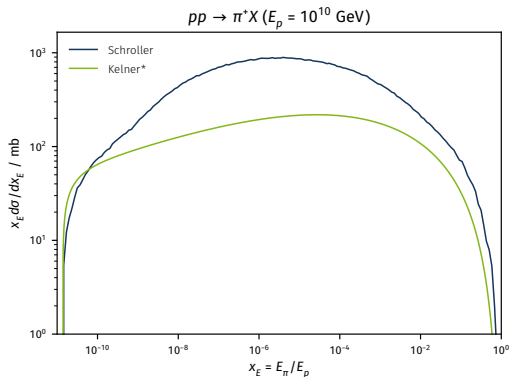


Monte Carlo Event Generation



Data Courtesy
M. Schroller

Differential Cross Sections Comparison



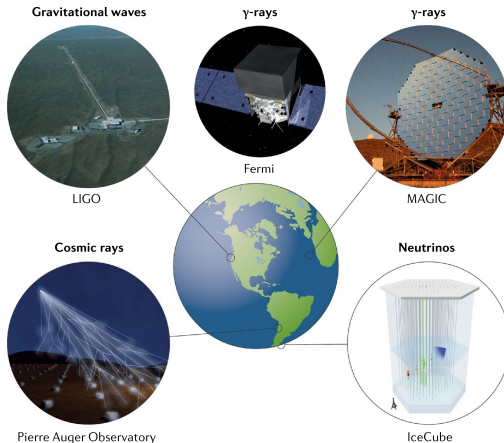
Derived Parametrization S. R. Kelner et al. *Phys. Rev. D* (2006)
Event Data (SIBYLL 2.3c) Courtesy M. Schroll



Summary & Conclusion

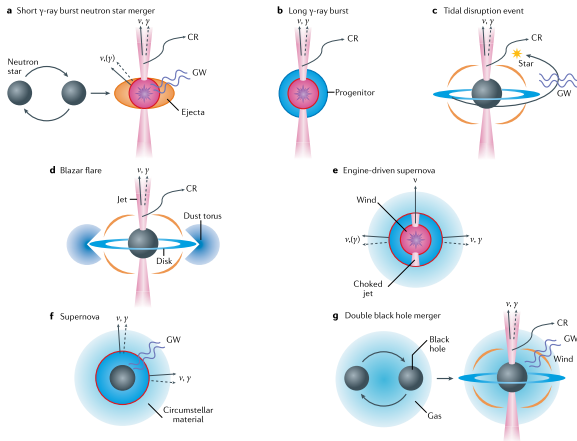
GitHub Repository: github.com/fritzali/bachelor

Messenger Types & Detection



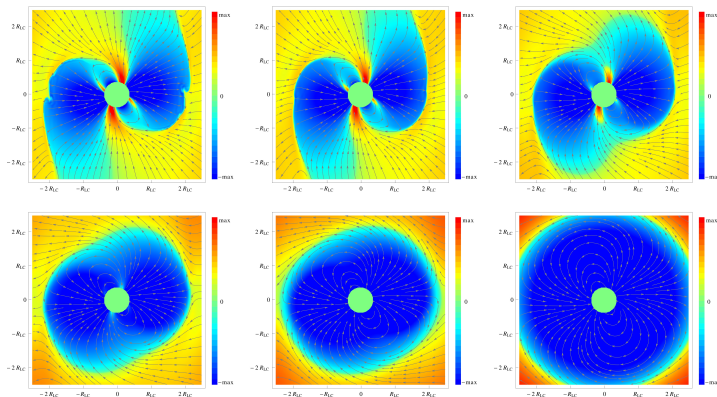
P. Mészáros et al.
Nat. Rev. Phys. (2019)

Source Regions & Events



P. Mészáros et al.
Nat. Rev. Phys. (2019)

Resistive Pulsar Magnetospheres



J. Li et al. *ApJ* (2012)

Global Luminosity Models

- Vacuum Dipole Radiation

$$L = \frac{2\mu^2\omega^4\sin^2\chi}{3c^2}$$

- Force Free Magnetosphere

$$L = \frac{\mu^2\omega^4(1 + \sin^2\chi)}{c^3}$$

Spindown Derivation

■ Magnetar Luminosity

$$L = K\omega^4 = -\dot{E}$$

■ Exponential Decay

$$E = E_0 \exp(-t/t_{sd}) \quad E = I\omega^2/2$$

$$\dot{E} = -E/t_{sd} \quad \dot{E} = I\omega\dot{\omega}$$

$$L_0 = K\omega_0^4 = I\omega_0^2/2t_{sd}$$

Spindown Derivation

- Spindown Time

$$t_{sd} = I/2K\omega_0^2$$

- Differential Equation

$$I\dot{\omega}_0 = -K\omega^3$$

- Separating Variables

$$\omega = \omega_0(1 + t/t_{sd})^{-1/2}$$

Acceleration Mechanisms

■ Wakefield Acceleration

$$E^M = \eta e \phi_{rc} = \eta e \psi_{rc} / R_{lc} = \eta e \mu_{ns} / R_{lc}^2$$

■ Diffusive Shock Acceleration

$$N = N_0 \zeta^k \quad E = E_0 \eta^k$$

$$\frac{\ln(N/N_0)}{\ln(E/E_0)} = \frac{\ln(\zeta)}{\ln(\eta)} \quad N = N_0 \left(\frac{E}{E_0} \right)^{\ln(\zeta)/\ln(\eta)} \quad \frac{dN}{dE} = \frac{N_0}{E_0} \left(\frac{E}{E_0} \right)^{\ln(\zeta)/\ln(\eta) - 1}$$

Hydrostatic Disk Equilibrium

- Gravitational Force

$$F = \frac{GM_{bh}m_p}{R_{dc}^2}$$

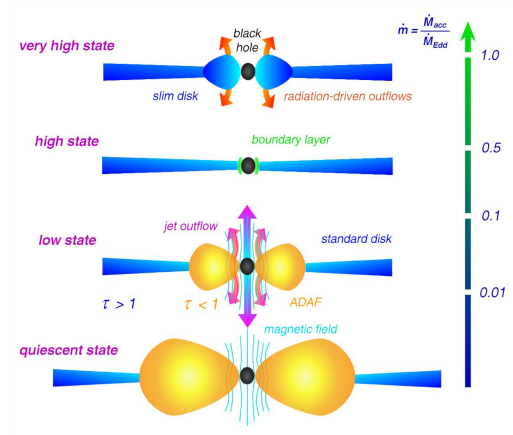
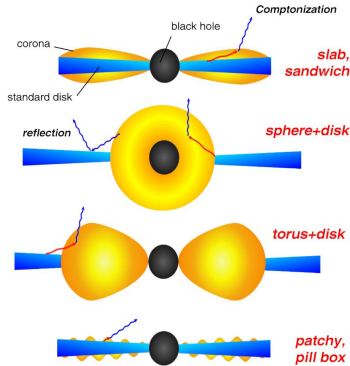
- Small Angle Approximation

$$\frac{d\rho}{dz} = -\rho\ddot{z} = -\frac{\rho GM_{bh}z}{u_{sn}^2 R_{dc}^3}$$

- Sound & Orbital Velocity

$$\rho = \rho_0 \exp(-z^2/2h^2) \quad h^2 = \frac{u_{sn}^2 R_{dc}^3}{GM_{bh}} = \frac{u_{sn}^2 R_{dc}^2}{v_{or}^2}$$

Accretion Disk & Corona



A. Müller Heidelberg University (2004)

Particle Interactions

■ Source Size Relation

$$R = \frac{P}{QB} \quad E < Ze\beta cBD$$

■ Mean Lifetime & Free Path

$$\lambda = (\kappa\sigma n)^{-1}$$

■ Decay & Dissipation Laws

$$P_x = 1 - \exp(-x/\lambda) \quad P_t = 1 - \exp(-t/\tau)$$

Parametrized Distributions

■ Hadron Scattering

$$\sigma_{hp} = H_h \ln^2(s/s_h) + P_h + R_h^1 (s_h/s)^{\eta_1} + R_h^2 (s_h/s)^{\eta_2}$$

■ Charm Cross Section

$$x_F = p_c/p_s \quad x_F \frac{d\sigma}{dx_F} = a x_F^b (1 - x_F^m)^n$$

$$a = a_1 \ln(E_p) - a_2 \quad b = b_1 - b_2 \ln(E_p) \quad n = n_1 - n_2 \ln(E_p)$$

Parametrized Distributions

■ Neutral Pion Distribution

$$F_{pp \rightarrow \pi} = 4E_p^{-1} \alpha B x_\pi^{\alpha-1} \left(\frac{y_\pi}{1 + r x_\pi^\alpha y_\pi} \right)^4 \left(y_\pi^{-1} + \frac{r(1 - 2x_\pi^\alpha)}{1 + r x_\pi^\alpha y_\pi} \right) \left(1 - \frac{m_\pi}{x_\pi E_p} \right)^{1/2}$$

$$y_\pi = 1 - x_\pi^\alpha \quad B = \tilde{B} + C \quad \alpha = \tilde{\alpha} / \sqrt{C} \quad r = \tilde{r} / \sqrt{C}$$

$$C = c_1 - c_2 \ln(E_p) + c_3 \ln^2(E_p)$$

Parametrized Distributions

■ Effective Three Body Decays

$$F_{h \rightarrow \nu} = B_{h \rightarrow \nu} E_h^{-1} D_h^{-1} \left(6b_h a_h^2 - 4a_h^3 - 12\lambda_h^2 a_h + 12\lambda_h^2 y - 6b_h y^2 + 4y^3 + 12\lambda_h^2 \ln((1-y)/\lambda_h) \right)$$

$$y = E_\nu / E_h \quad \lambda_h = s_h / m_h^2 \quad a_h = 1 - \lambda_h \quad b_h = 1 - 2\lambda_h$$

$$D_h = 1 - 8\lambda_h - 12\lambda_h^2 \ln(\lambda_h) + 8\lambda_h^3 - \lambda_h^4$$